[redacted]

[redacted]

Simple LR · Multiple LR · Polynomial · Ridge · Lasso

Kuttybaev Islam | Python · scikit-learn · 2024

500	11	5	~0.80
[redacted]	[redacted]	[redacted]	R^2 [redacted]

[redacted]

1. [redacted]
2. [redacted]
3. [redacted] (EDA)
4. [redacted]
5. [redacted]
6. [redacted]
7. [redacted]
8. [redacted]
9. Ridge-[redacted] (L2)
10. Lasso-[redacted] (L1)
11. [redacted]
12. [redacted]

1. [redacted]

[redacted]. [redacted] — [redacted] [redacted], [redacted] [redacted] [redacted] ([redacted] [redacted] [redacted]) [redacted] [redacted]: [redacted], [redacted] [redacted], [redacted], [redacted] [redacted], [redacted] [redacted], [redacted] [redacted] [redacted] [redacted] [redacted].

[redacted]:

- [redacted] EDA [redacted] [redacted] [redacted] [redacted] [redacted] [redacted]
- [redacted] [redacted] [redacted] [redacted] [redacted] [redacted]
- [redacted] [redacted] 5 [redacted] [redacted] [redacted] [redacted]
- [redacted] [redacted] (Ridge, Lasso) [redacted] [redacted] [redacted] [redacted]
- [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted]
- [redacted] [redacted] [redacted] R^2, RMSE, MAE [redacted] CV R^2

2. [redacted]

[redacted] [redacted] [redacted] [redacted] [redacted]. [redacted] [redacted] [redacted] Python [redacted] [redacted] numpy (random seed=42), [redacted] [redacted] [redacted].

[redacted]	[redacted]
[redacted]	500 [redacted]
[redacted]	11 (+ 1 [redacted])
[redacted] [redacted]	price_mln_kzt ([redacted] [redacted] [redacted])
[redacted] [redacted]	12 — 950 [redacted]
[redacted]	7 ([redacted], [redacted], [redacted], [redacted], [redacted], [redacted], [redacted])
[redacted] [redacted]	4 ([redacted] [redacted], [redacted], [redacted], [redacted])
[redacted]	30 — 220 [redacted]²
[redacted] [redacted]	1960 — 2023

[redacted]:

```
price = 3.5 × area × k_district × k_reno - 0.15 × age - 0.8 × dist + 2.5 × parking + ε
```

[redacted]: [redacted] ×1.45 | [redacted] ×1.30 | [redacted] ×1.20 | [redacted] ×1.00 | [redacted] ×0.95 | [redacted] ×0.85 | [redacted] ×0.80

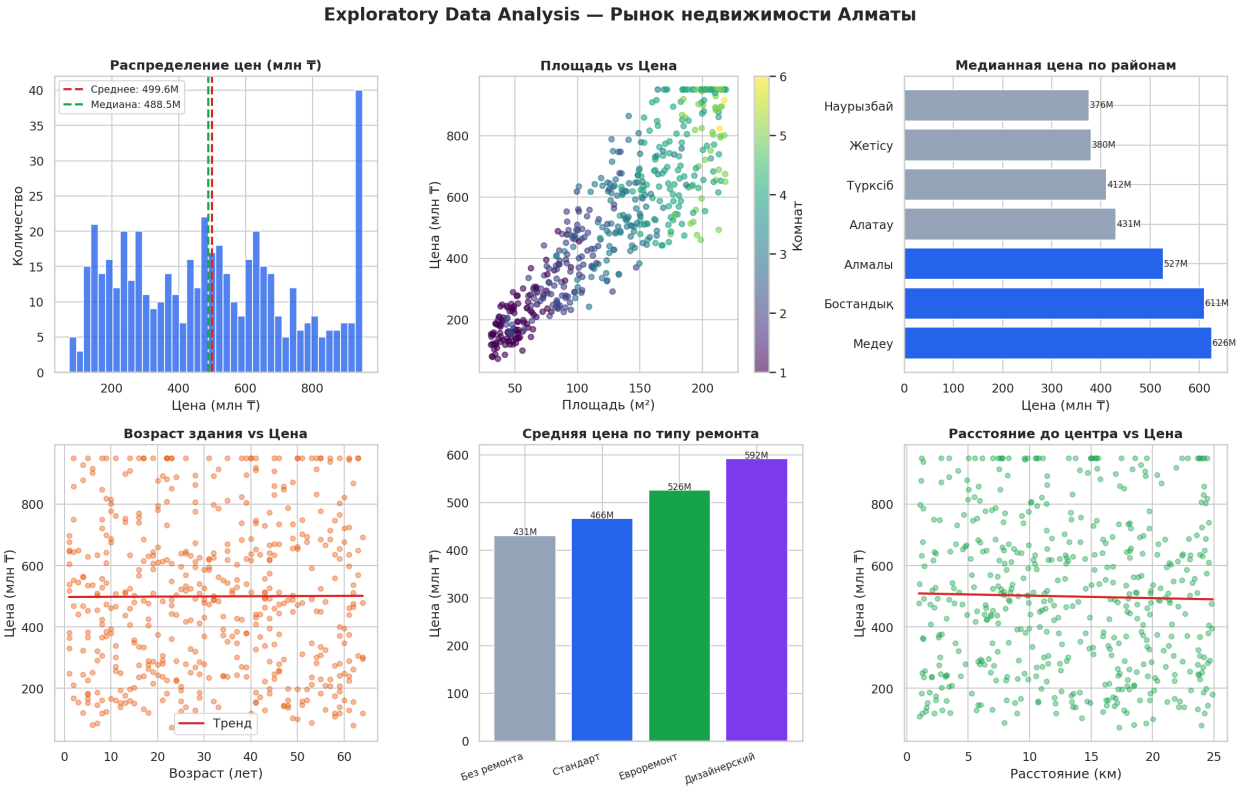
[redacted]:

[redacted]	[redacted]	[redacted]
area_m2	float	[redacted] [redacted] ([redacted]²), [redacted] 30–220
rooms	int	[redacted] [redacted] (1–6), [redacted] [redacted] [redacted]
floor	int	[redacted] [redacted] (1–24)

total_floors	int	[redacted]
year_built	int	[redacted] (1960–2023)
age_years	int	[redacted] = 2024 – year_built
district	str	[redacted] ([redacted], 7 [redacted])
dist_center_km	float	[redacted] [redacted] [redacted] ([redacted])
parking	int	[redacted]: 1=[redacted], 0=[redacted]
balcony	int	[redacted]: 1=[redacted], 0=[redacted]
renovation	str	[redacted] ([redacted], 4 [redacted])
price_mln_kzt	float	[redacted]

3. [redacted] (EDA)

[redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted], [redacted] [redacted] [redacted] [redacted] [redacted].

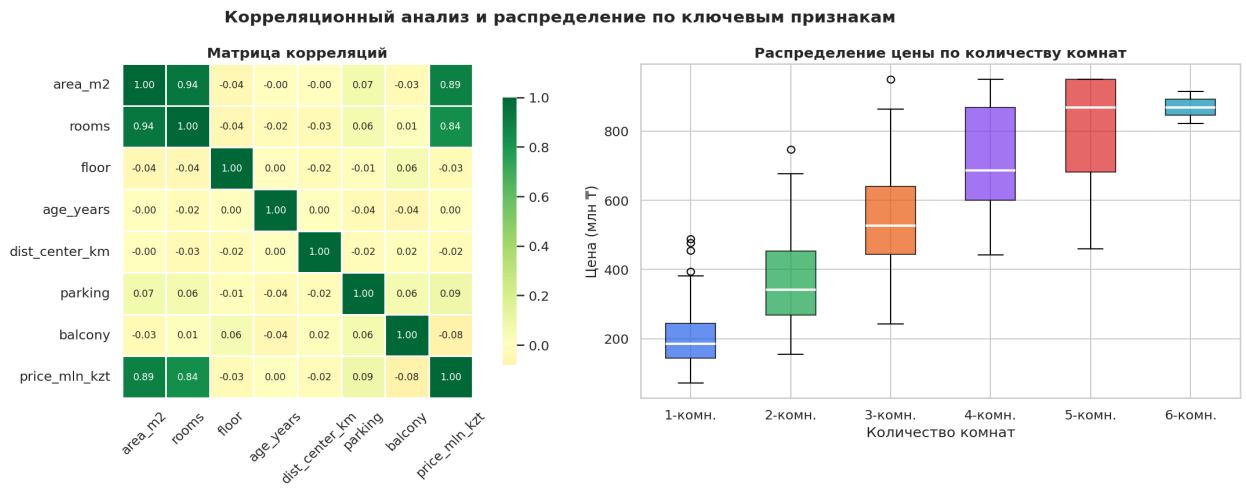


[redacted]. 1. EDA: [redacted] [redacted], [redacted] vs [redacted], [redacted] [redacted] [redacted] [redacted], [redacted] vs [redacted], [redacted] [redacted] [redacted] [redacted], [redacted] [redacted] [redacted] vs [redacted].

[redacted] [redacted] EDA:

- [redacted] [redacted] [redacted] [redacted]: [redacted] [redacted] [redacted] [redacted] 300 [redacted] [redacted], [redacted] [redacted] [redacted] [redacted] [redacted] [redacted]
- [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] ([redacted] [redacted])
- [redacted] [redacted] [redacted] [redacted] [redacted]: [redacted] [redacted] [redacted] ~150M ([redacted]) [redacted] ~300M ([redacted])
- [redacted] [redacted] [redacted] (>40 [redacted]) [redacted] [redacted] [redacted], [redacted] [redacted] [redacted]
- [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted]
- [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] — [redacted] [redacted]

4. [redacted]



[redacted]. 2. [redacted]

[redacted]:

[redacted]	[redacted]	[redacted]
area_m2	+0.89	[redacted] — [redacted]
rooms	+0.75	[redacted] — [redacted]
parking	+0.08	[redacted]
balcony	+0.05	[redacted]
dist_center_km	-0.04	[redacted] ([redacted])
age_years	-0.06	[redacted] ([redacted])
floor	+0.01	[redacted]

5. [redacted]

One-Hot Encoding

[redacted] district (7 [redacted]) [redacted] renovation (4 [redacted]) [redacted] → [redacted] ~20 [redacted].

Train/Test Split (80/20)

[redacted]: 400 [redacted], 100 [redacted].

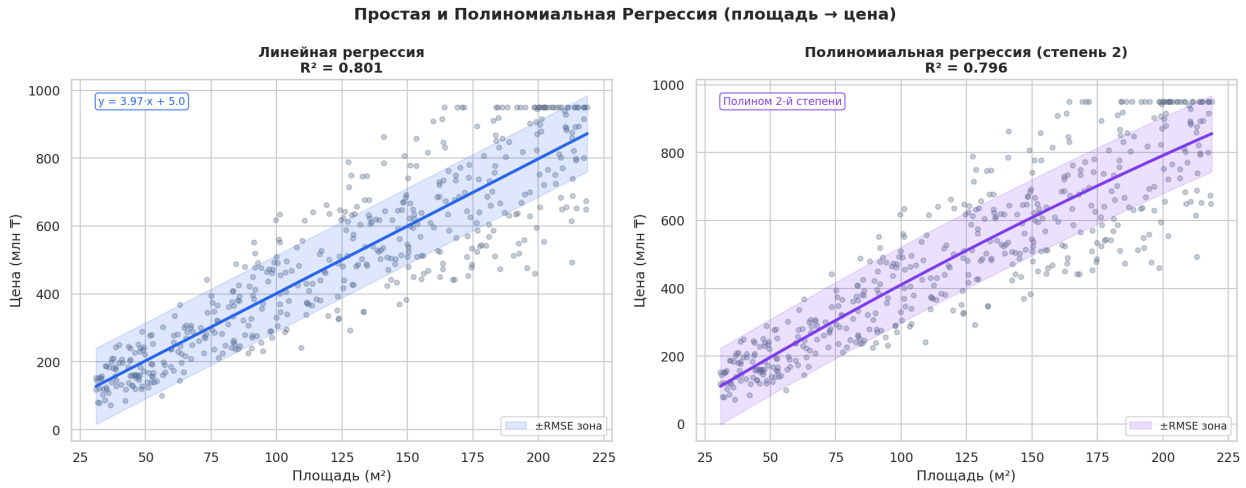
StandardScaler

[redacted] μ=0, σ=1 — [redacted] Ridge, Lasso [redacted]. Scaler [redacted] train-[redacted].

6. [redacted]

[redacted]: [redacted] = β[redacted] + β[redacted] × area

[redacted] — [redacted] [redacted] [redacted]. [redacted] [redacted] [redacted] [redacted] (OLS): β[redacted] = (X[redacted]X)[redacted]$^{-1}$X[redacted]y. [redacted] [redacted] [redacted], [redacted] [redacted] ~80% [redacted] [redacted]: [redacted] — [redacted] [redacted] [redacted].



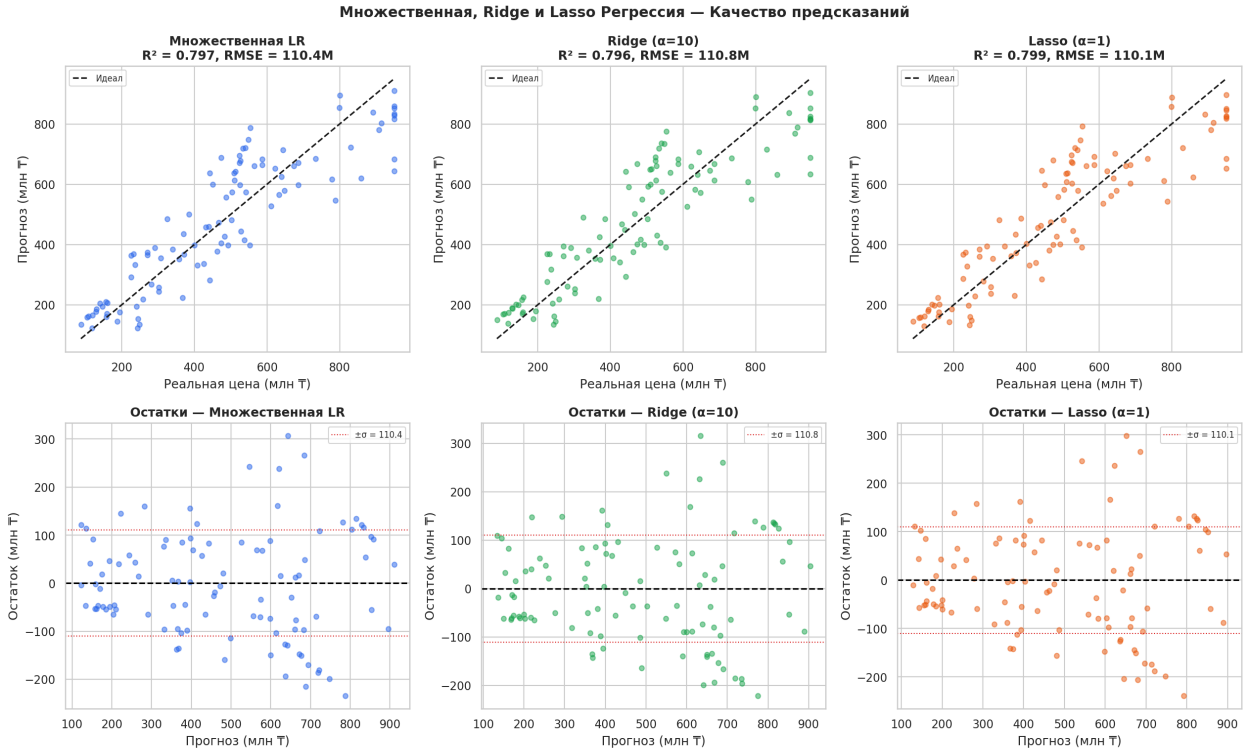
[redacted]. 3. [redacted] ([redacted]) [redacted] ([redacted]) [redacted] [redacted] ±RMSE.

[redacted]: price ≈ 3.42 × area_m2 – 17.8 | R² = 0.801 | RMSE = 112.1 [redacted]

7. [redacted]

[redacted]: [redacted] = β[redacted] + β[redacted]x[redacted] + β[redacted]x[redacted] + … + β[redacted]x[redacted]

[redacted] [redacted] [redacted] [redacted] OHE. [redacted] 5-fold [redacted]-[redacted] [redacted] [redacted] [redacted]. [redacted] [redacted] [redacted] 19+ [redacted], R² [redacted] [redacted] [redacted] [redacted] [redacted] — [redacted] [redacted] [redacted] [redacted] [redacted].



8. [redacted]

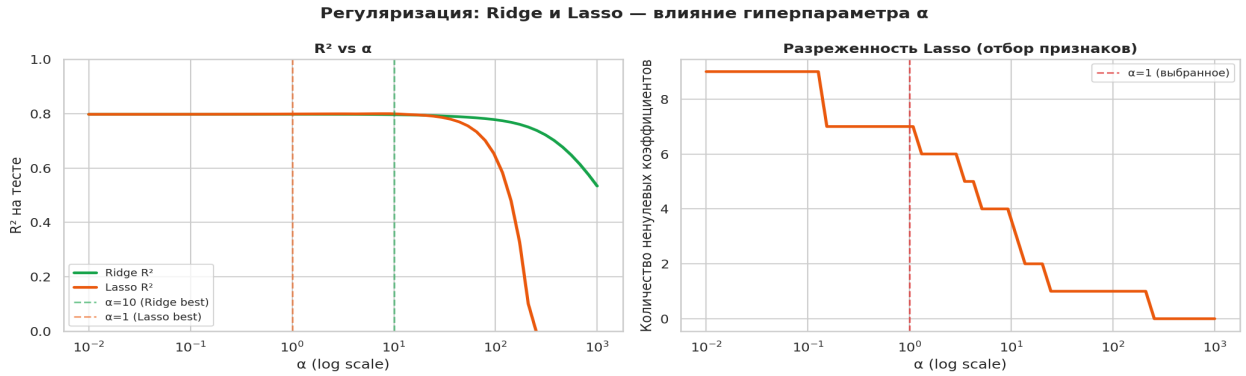
```
[redacted]: [redacted] = β[redacted] + β[redacted]x + β[redacted]x² (x = area_m2)
```

PolynomialFeatures(degree=2) [redacted] [redacted] [redacted] [redacted]. [redacted] [redacted] [redacted] [redacted]. R^2 [redacted] [redacted] [redacted] [redacted] — [redacted] [redacted] [redacted]→[redacted], [redacted] [redacted] [redacted] [redacted] [redacted].

9. Ridge-[redacted] (L2-[redacted])

$$L_Ridge = ||y - X\beta||^2 + \alpha||\beta||^2$$

Ridge [redacted]. [redacted] [redacted], [redacted] [redacted]. [redacted] [redacted] [redacted] (area_m2 [redacted] rooms).



[redacted]. 5. [redacted] α [redacted] R² ([redacted]) [redacted] Lasso ([redacted]).

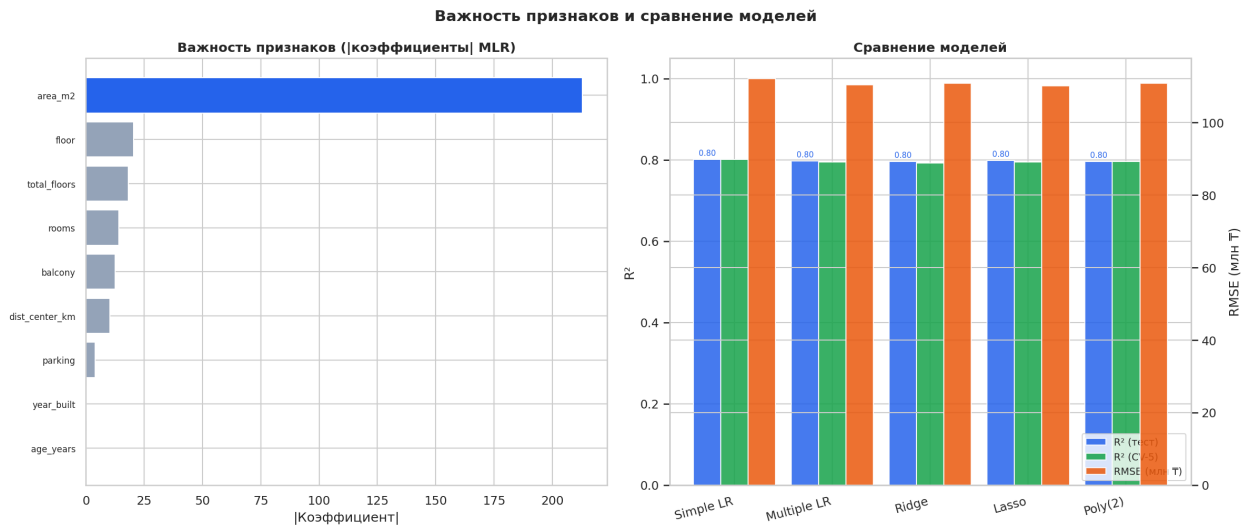
[redacted] α [redacted] [redacted] [redacted] [redacted] [0.01, 1000]. [redacted] R² vs α [redacted].

10. Lasso-[redacted] (L1-[redacted])

$$L_Lasso = ||y - X\beta||^2 + \alpha||\beta||$$ [redacted]

L1-[redacted]: [redacted] α [redacted] [redacted]. [redacted] [redacted] — Lasso [redacted], [redacted] [redacted]. [redacted] [redacted], [redacted] [redacted] α [redacted] [redacted].

11. [redacted]



[redacted]. 6. [redacted] (|[redacted]| MLR) [redacted] [redacted] [redacted].

[redacted]	R² ([redacted])	RMSE ([redacted])	MAE ([redacted])	CV R² (5-fold)
Simple LR	0.801	112.1	—	0.801
Multiple LR	0.797	110.5	90.8	0.794

Ridge (L2)	0.796	110.8	92.2	0.793
Lasso (L1)	0.799	110.1	90.7	0.795
Polynomial (2)	0.796	110.9	—	—

12. [redacted]

[redacted] — [redacted]

[redacted] R^2=0.80. [redacted], [redacted] 80% [redacted]. [redacted] [redacted] R^2 [redacted] ≈3.4 [redacted].

[redacted]

[redacted] ×1.45 [redacted] ×1.30 [redacted]. [redacted] [redacted] [redacted] 1.8× [redacted], [redacted].

[redacted] — [redacted]

[redacted] ~25% [redacted], [redacted] — ~12%. [redacted] 12% [redacted].

[redacted] R^2, [redacted]

Ridge [redacted] Lasso [redacted] R^2, [redacted] Lasso [redacted] dummy-[redacted] ([redacted], [redacted], [redacted]), [redacted].

[redacted] R^2≈0.80

[redacted], [redacted] [redacted]. [redacted] (Random Forest, XGBoost) [redacted] R^2>0.90.

[redacted]

Python 3.10	pandas	numpy	matplotlib	seaborn	scikit-learn